MedVES: Visual Evidence Scaling at Test Time for VLMs in Medical VQA

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Abstract

Medical vision-language models (VLMs) hold significant promise for clinical image understanding, but scaling them at training time is limited by data privacy constraints and the cost and scarcity of expert annotations. Test-time scaling (TTS) is therefore a practical alternative, as it can improve performance without retraining. However, existing TTS methods are designed mainly for LLMs, where extra inference compute is spent on language-space reasoning. This paradigm does not transfer directly to medical visual reasoning, where generating more text does not yield new visual findings. Instead, progress requires iteratively seeking and integrating fine-grained, localized visual evidence from different image regions. We introduce MedVES, a training-free framework for visual evidence scaling at test time for frozen VLMs in medical VQA. MedVES casts test-time scaling as iterative visual evidence seeking rather than extended language-space reasoning. Its core component is a Visual Overlay State (VOS), a bounded visual state representation that externalizes accumulated evidence on the original image while preserving global context. Built on VOS, MedVES performs iterative visual evidence acquisition with adaptive stopping, enabling test-time compute to scale with question complexity. Across five medical VQA benchmarks covering diverse imaging modalities, a 4B MedVES model achieves strong average performance at its scale and is competitive with, or better than, several models with 8× more parameters, without any additional training. Code is available.

CCS Concepts

• **Information systems** → **Multimedia information systems**; • **Applied computing** → **Health care information systems**; • **Computing methodologies** → *Computer vision.*

Keywords

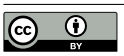
vision-language models, medical visual question answering, test-time scaling, multimodal reasoning

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1 Introduction

Recent advances in foundation models have been driven largely by scaling training-time resources such as model size, data, and optimization compute. However, intelligent problem solving is shaped not only by training but also by the computation spent at inference: humans often reason better by thinking longer, examining evidence, and checking intermediate conclusions before responding. This intuition has led to growing interest in test-time scaling (TTS), which

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improves model performance by increasing inference-time computation rather than retraining models or expanding datasets. This paradigm is particularly attractive in medicine. Despite the strong potential of VLMs for medical image understanding and multimodal reasoning [6, 28, 35, 44], training-time scaling is severely limited by patient data privacy constraints and the expense and scarcity of high-quality expert annotations for supervised learning, making conventional scaling strategies difficult to apply in practice [22].

In large language models (LLMs), TTS is typically realized through additional language-space computation, such as sampling and selecting among multiple reasoning paths [49, 50] or searching over intermediate thoughts [9, 56]. These approaches are effective in text-only settings because intermediate reasoning states can be naturally serialized as tokens, allowing later steps to refine, revise, or verify earlier ones. This makes it possible for the model to effectively "think longer" before committing to an answer [5, 24, 57].

However, this paradigm does not transfer directly to VLMs. **More generated text tokens do not yield new visual findings**. Instead, they merely elaborate on the model's initial impression rather than uncovering missing visual information. Effective test-time scaling for VLMs should therefore actively attend to different image regions and seek new, non-redundant visual evidence that can support answer generation. This introduces three challenges: (1) **Evidence seeking**: how to guide the model, conditioned on what has already been observed, to discover new and non-redundant visual evidence; (2) **State representation**: how to encode the accumulated visual evidence into an efficient, reusable, and spatially grounded state; (3) **Scaling efficiency**: how to improve performance through additional evidence-seeking steps while keeping the associated token cost under control.

We address these challenges by presenting **MedVES**, a training-free test-time scaling framework for VLMs in medical VQA. The central idea of MedVES is to cast test-time scaling as an iterative process of *visual evidence seeking* prior to answer generation, rather than as extended language-space reasoning. Concretely, given an image-question pair, a *frozen* VLM is repeatedly prompted to attend to different image regions, propose discriminative visual evidence, and select the next evidence-seeking step from a compact action vocabulary that provides one practical instantiation of sequential evidence acquisition. In our current implementation, this vocabulary includes symmetry verification, adjacent-context checking, and subregion refinement. We keep this action vocabulary compact and generic so that the framework remains training-free and broadly applicable across medical VQA settings. To support this process, MedVES externalizes the intermediate reasoning state into a primarily *visual* state rather than a purely textual one. We propose the **Visual Overlay State** (VOS), in which accumulated evidence is rendered as lightweight bounding box overlays directly onto the original image. By maintaining all accepted evidence on the full original images, VOS provides a fixed-size, spatially grounded, and reusable state representation that preserves global context by construction. Built on VOS, MedVES performs state-conditioned evidence acquisition: each accepted evidence step updates the visual state, and the subsequent query is conditioned on the updated state to seek complementary evidence. After sufficient evidence has been gathered, the model produces the final answer conditioned on the accumulated visual state. Finally, to make this scaling process efficient, MedVES adopts an adaptive evidence budget with an early-exit mechanism, allowing the model to terminate evidence collection once sufficient evidence has been gathered, allocate compute proportional to question complexity, and prevent noise accumulation from redundant steps. Our goal is therefore not to minimize inference cost absolutely, but to improve the efficiency-performance trade-off relative to larger fixed-budget settings. We evaluate MedVES comprehensively on five medical VQA benchmarks spanning diverse imaging modalities. Under controlled comparisons using the same frozen backbone, MedVES improves average accuracy on OmniMedVQA from 72.0% to 80.5% and substantially outperforms representative training-free TTS baselines. Overall, the resulting 4B model achieves strong average performance and is competitive with, or better than, several models with up to 8× more parameters, without any additional training.

Our main contributions are as follows:

- We present, to the best of our knowledge, the first training-free test-time scaling for VLMs in medical VQA, and introduce **MedVES**, an inference framework that requires no retraining or additional annotated data.
- We propose **Visual Overlay State (VOS)**, a unified visual state representation that enables iterative evidence seeking and adaptive budget allocation for inference-time scaling. VOS preserves global context, grounds each evidence-seeking step in accumulated visual evidence, and supports adaptive compute allocation during inference.
- Across five medical VQA benchmarks, our 4B MedVES model achieves strong average performance among the evaluated models at a comparable scale and remains competitive with substantially larger models, even outperforming several of them. Extensive ablations further validate the contribution of each component.

2 Related Work

2.1 General Vision-Language Models

General vision-language models (VLMs) combine visual encoders with large language models to support multimodal understanding and generation. Foundational models such as Flamingo [2], BLIP-2 [30], and LLaVA [34] show that strong multimodal capabilities can be achieved by aligning pretrained vision backbones with large language models through lightweight connectors and instruction tuning. More recent open VLM families, including InternVL [7, 8], DeepSeek-VL2 [52], Qwen3-VL [4], and Gemma 3 [47], further improve general-purpose multimodal reasoning through larger-scale training, longer context, and more efficient architectures. Despite this progress, most prior work advances VLMs through model scaling, architectural design, or additional supervision. Training-free test-time scaling with explicit reusable visual states has remained largely unexplored.

2.2 Medical Vision Language Models

Medical vision language models integrate large language models with visual perception to analyze and interpret medical images, enabling clinical reasoning and decision support [6, 28, 38, 51, 58].

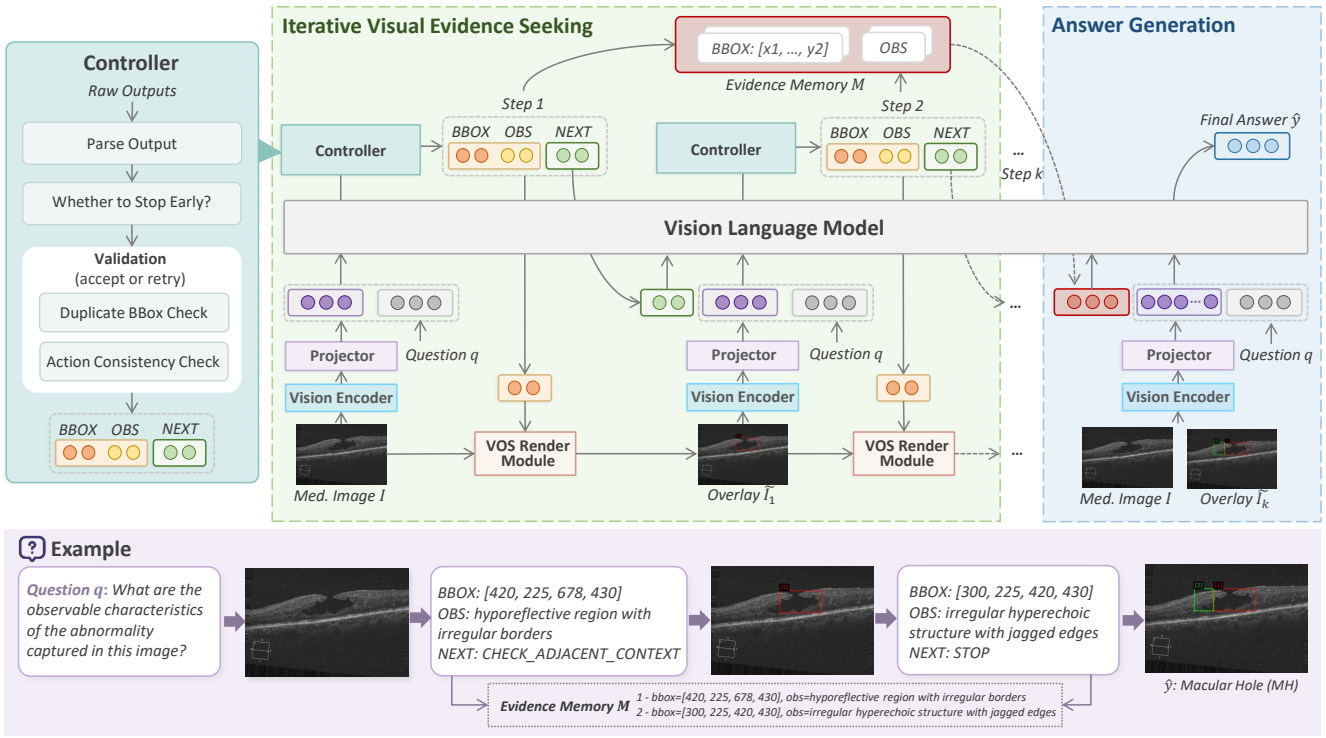


Figure 1: Framework overview of MedVES. Given a medical image and a question, MedVES iteratively queries a frozen VLM to seek discriminative visual evidence, accumulates accepted evidence in a Visual Overlay State (VOS), and adaptively terminates evidence collection via an early-exit mechanism before producing the final answer. The retinal optical coherence tomography (OCT) image (© O. S. Naren, *Retinal OCT-C8*) was sourced via the OmniMedVQA dataset [15]. Used under CC BY 4.0.

Early systems such as Med-Flamingo [38] and LLaVA-Med [28] adapted general VLMs to the medical domain through interleaved image-text supervision, demonstrating strong generative and VQA performance under few-shot settings. MedVInT [58] aligned visual inputs with language models using curated VQA pairs, while RadFM extended this paradigm to both 2D and 3D radiologic imaging [51]. Huatuo-Vision [6], BiomedGPT [36], MedGemma [44], and Lingshu [26] further advanced large-scale multimodal medical reasoning. More recent work has expanded medical VLMs along several dimensions, including multi-image reasoning [55], retrieval-augmented generation [53], agent-style visual reasoning [13], 3D architectural design [54], diffusion-based modeling [11], and reinforcement learning for more interpretable and reliable reasoning [42]. Despite these advances, most prior work has focused on model adaptation, architecture, retrieval, or training-based reasoning improvements, whereas training-free test-time scaling has been largely unexplored.

2.3 Test-Time Scaling in General LLMs/VLMs

OpenAI-o1 [17] popularized test-time scaling (TTS) with a long-reasoning mechanism that performs implicit reflection and revision during generation, producing substantially longer chains of thought and yielding strong gains in challenging domains such as math [31] and coding [18]. Subsequent studies show that such long-reasoning behaviors can be strengthened by post-training, either via reinforcement learning on structured data [12, 48] or supervised fine-tuning on long-reasoning corpora [12, 16, 39, 41]. In parallel, a complementary family of training-free TTS baselines improves reliability by sampling and aggregating multiple candidates, including Best-of-N selection [49], majority voting [10], and search-based decoding, such as Monte Carlo Tree Search (MCTS) [9]. More recently, "R1-style" TTS has been extended from language-only settings to vision-language models through grounded visual operations [46]. Related work has also explored grounded visual reasoning via reinforcement learning or external tools [25, 40, 43, 45, 59], often relying on additional training or specialized supervision. Ophiuchus [19] and MedVR [21] learn medical tool-use or visual-reasoning policies through additional training or reinforcement learning, while RTWI [29] filters and votes over noisy image-text thinking traces, and TTSP [20] samples and filters exploratory perception and tool-use traces. In contrast, we study test-time scaling in a training-free setting with an explicit cumulative visual state.

3 Method

We aim to enable test-time scaling for VLMs in medical VQA by casting inference as iterative visual evidence seeking rather than extended language-space reasoning. To this end, we propose **MedVES**, a training-free framework built on a frozen VLM backbone. MedVES comprises three components: (1) a **Visual Overlay State** (VOS) that accumulates accepted evidence as lightweight bounding-box

Algorithm 1 MedVES

Require: Image I, question q, evidence budget K, retry budget R, IoU threshold δ
Ensure: Predicted answer $\hat{y}$

$M \leftarrow \emptyset$ ▷ accepted evidence memory: tuples $(\mathbf{b}, o)$
$\tilde{I} \leftarrow I$ ▷ initial visual overlay state
$a \leftarrow$ INIT ▷ initial control action
for $k \leftarrow 1$ **to** K **do**
  accepted $\leftarrow$ **false**
  for $r \leftarrow 1$ **to** R **do**
    $s \leftarrow \textsc{QueryEvidence}(I, \tilde{I}, q, a)$ ▷ action-conditioned evidence-seeking
    Parse $s \rightarrow (\mathbf{b}, o, n)$
    if $\mathbf{b}$ = NONE **and** o = NONE **and** n = STOP **then**
      goto FinalAnswer
    end if
    if $\textsc{DuplicateBox}(\mathbf{b}, M, \delta, a)$ **then** ▷ reject proposal
      continue
    end if
    if not $\textsc{ActionConsistent}(\mathbf{b}, M, a)$ **then**
      continue
    end if
    $M \leftarrow M \cup \{(\mathbf{b}, o)\}$ ▷ accept evidence
    $\tilde{I} \leftarrow \textsc{Render}(I, B_k)$ ▷ cumulative overlay on the original image
    $a \leftarrow n$ ▷ next-step control action
    accepted $\leftarrow$ **true**
    if n = STOP **then**
      goto FinalAnswer
    end if
    break
  end for
  if not accepted **then**
    break ▷ retry budget exhausted
  end if
end for

FinalAnswer:
$\hat{y} \leftarrow \textsc{QueryAnswer}(I, \tilde{I}, q, M)$ ▷ condition on the accumulated evidence memory
return $\hat{y}$

overlays on the original image, providing a reusable and spatially grounded visual state that preserves global context throughout inference (Sec. 3.2); (2) an **Iterative Visual Evidence-Seeking** procedure that repeatedly queries the frozen VLM to propose, verify, and update discriminative visual evidence conditioned on the current visual state (Sec. 3.3); and (3) an **Adaptive Evidence Budget** that scales test-time compute by collecting multiple pieces of evidence while using early exit to terminate once sufficient evidence has been gathered (Sec. 3.4).

3.1 Framework Overview

Given a medical image I and a medical question q, the goal is to predict an answer $\hat{y}$. Standard single-pass inference feeds (I, q) directly to a VLM backbone for medical VQA and generates an answer in a single step, implicitly assuming that all visually relevant evidence is captured and used in a global image encoding. This assumption often breaks down in medical VQA: clinically decisive features, such as fine lesion boundaries, focal asymmetries, and faint density changes, are often spatially sparse and may be suppressed in a holistic feature representation. Without an explicit mechanism to direct attention toward such regions, the model may answer before it has identified the evidence it needs. We therefore argue that, when sufficient test-time compute is available, improving the performance of existing VLM backbones on medical VQA requires a structured evidence-seeking phase before answering, which can be realized at test time as a form of explicit visual reasoning.

To operationalize this evidence-seeking phase, we introduce a test-time scaling controller that guides a *frozen* VLM $\mathcal{M}$ to iteratively collect structured visual evidence prior to answering. At each step k, the controller queries $\mathcal{M}$ with the medical image, the current visual state, the question, and a compact control signal indicating what type of evidence to seek next, and obtains an *evidence tuple*

$$e_k = (\mathbf{b}_k,\ o_k,\ n_k), \qquad k = 1, \ldots, K, \tag{1}$$

where $\mathbf{b}_k = [x_1, y_1, x_2, y_2]$ is a 2D bounding box with coordinates normalized to $[0, 1000]$, o_k (OBS) is a concise, purely visual observation within that region, and n_k (NEXT) is a discrete control action that specifies how the subsequent evidence-seeking step should proceed. For $n_k \neq$ STOP, the predicted NEXT field from step k is fed back to condition the query at step $k+1$, forming an iterative action-conditioned evidence-seeking loop. After collecting up to K accepted evidence tuples, the controller issues a final query conditioned on the original image, the cumulative visual overlay, the medical question, and the accumulated evidence memory to produce the answer $\hat{y}$.

3.2 Visual Overlay State

A central challenge in iterative evidence-seeking is maintaining a state representation that remains both informative and bounded: it should preserve the information needed from prior steps without growing in a way that weakens focus on the most recently discovered evidence. A naive solution is to append each exchange to a dialogue history; however, this leads to rapid token explosion as K increases. We instead externalize the spatial trace of reasoning into a *Visual Overlay State* (VOS): a persistent, image-based structure that records which regions have been examined so far directly on the original image. This design keeps the visual state bounded, since the model always receives a fixed-size image overlay regardless of K, while still making prior search trajectories spatially explicit. Because prior search trajectories are rendered visually, the model can exploit its spatial attention mechanisms to directly perceive what has been examined and where. This yields an approximately state-centric interface: at each step, the model is primarily guided by the current visual state $(I, \tilde{I}_{k-1})$. Conceptually, VOS serves as a bounded state abstraction for sequential visual evidence acquisition. At every evidence step, the model receives both the original image I and the overlay $\tilde{I}_{k-1}$, so the overlay does not replace or occlude the underlying pixels. VOS records *where* the model has inspected, while the corresponding OBS entries record *what* was observed; an unboxed region denotes a region not yet inspected by the controller.

Memory state. We define the accepted evidence memory after step k as

$$M_k = \{(\mathbf{b}_1, o_1), \ldots, (\mathbf{b}_k, o_k)\}, \qquad M_0 = \emptyset, \tag{2}$$

which records the spatial location and visual observation of each accepted evidence tuple. The NEXT field is intentionally excluded from memory, because it is a forward-looking directive used only to guide the subsequent evidence-seeking step.

Cumulative rendering. The visual overlay state is constructed from the spatial projection of memory rather than from the full

symbolic state. Let

$$B_k = \{\mathbf{b} \mid (\mathbf{b}, o) \in M_k\}. \tag{3}$$

We then define

$$\tilde{I}_k = \textsc{Render}(I, B_k), \tag{4}$$

where $\textsc{Render}(\cdot)$ cumulatively overlays all accepted bounding boxes in B_k onto the original image I. The resulting overlay explicitly indicates which regions have been inspected so far and serves as the bounded visual state for the next evidence-seeking step.

3.3 Iterative Visual Evidence Seeking

Action-conditioned evidence-seeking. We initialize the control action as `INIT`, and after each accepted step, use the predicted `NEXT` field to condition the subsequent evidence-seeking query. At step k, the controller queries the model with the original image I, the latest overlay image $\tilde{I}_{k-1}$, the question q, and a compact control signal indicating the intended type of next evidence. For later steps, `VERIFY_SYMMETRY`, `CHECK_ADJACENT_CONTEXT`, and `REFINE_SUBREGION` act as soft prompting cues that encourage a distinct comparison region, complementary context, or a finer-scale region, respectively. The model is required to output exactly three fields, `BBOX`, `OBS`, and `NEXT`, which are then parsed by the controller.

Evidence proposal and memory update. Given the current state, the model outputs a structured evidence tuple $(\mathbf{b}_k, o_k, n_k)$. The controller first parses the response into the required `BBOX/OBS/NEXT` format. If parsing fails, the controller issues a retry instruction without updating the state. Parsing fails when a required field is missing, `BBOX` cannot be parsed into four coordinates, or `NEXT` is outside the predefined action set. Otherwise, the proposed box is validated before acceptance. A candidate is rejected if its bounding box overlaps with any previously accepted evidence beyond an IoU threshold, preventing the model from repeatedly attending to the same region. In addition, the controller may reject proposals that are inconsistent with the requested action type (e.g., a non-adjacent region when the requested action is `CHECK_ADJACENT_CONTEXT`). Rejected proposals trigger a retry instruction that explicitly asks for a new box elsewhere. An accepted tuple triggers three updates. First, the controller appends $(\mathbf{b}_k, o_k)$ to the memory state M_k. Second, it renders the updated Visual Overlay State $\tilde{I}_k = \textsc{Render}(I, B_k)$ by cumulatively drawing all accepted bounding boxes on the original image. Third, it sets the next control action for the following step to $a_k = n_k$. This produces a closed loop in which each accepted step not only adds evidence to memory but also proposes how the next step should refine, contextualize, or verify that evidence. Accordingly, the intermediate tuples are best viewed as operational states for improving answer generation.

3.4 Adaptive Evidence Budget

The iterative evidence-seeking loop raises a key question: when should the model stop collecting evidence? The answer is inherently question-dependent. Some questions can be resolved from a small number of observations, while others require examining multiple candidate regions before sufficient evidence is available. A fixed budget K applies the same evidence-collection cost to all questions, which is inefficient: conservative budgets may stop too early, while large budgets improve accuracy but incur higher token cost.

Inference-time scaling with evidence steps. We study this trade-off by varying the fixed evidence budget. Figure 2 shows that increasing K consistently improves answer accuracy, confirming that iterative evidence seeking provides useful test-time scaling beyond single-pass inference. However, these gains come with a rapid increase in token usage, making larger fixed budgets increasingly expensive.

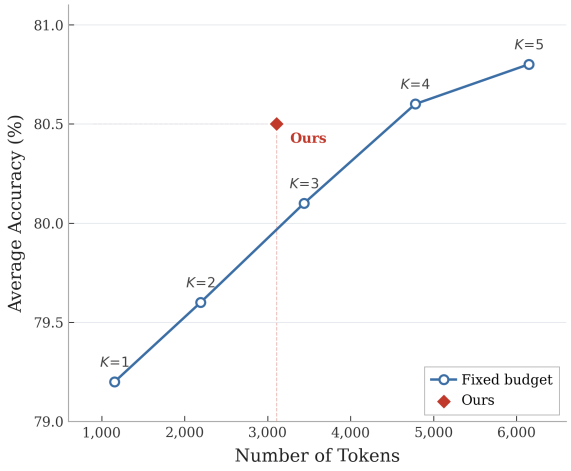


Figure 2: Inference-time scaling under different budgets. Accuracy increases steadily with budget K, while adaptive early exit achieves comparable accuracy with fewer tokens.

Adaptive early exit. To avoid spending unnecessary test-time compute on questions that can already be answered from the currently collected evidence, we equip the controller with an adaptive early-exit mechanism. Rather than imposing a fixed evidence budget for every example, we allow the frozen VLM itself to signal when further localization is unnecessary. Concretely, during evidence collection, the model may output `BBOX: NONE, OBS: NONE, NEXT: STOP` to indicate that no additional region needs to be localized. In this way, the stopping decision is delegated to the model's own reasoning over the question, the original image, the current overlay state, and the evidence accumulated so far. This yields question-adaptive compute allocation: simpler cases can terminate earlier, whereas more challenging cases can continue gathering evidence up to the maximum budget. As shown in Table 6 and Figure 2, this adaptive strategy achieves a substantially better efficiency-performance trade-off than fixed-budget inference, approaching the accuracy of larger fixed-budget settings at much lower cost.

4 Experiments

4.1 Experimental Setup

Implementation Details. Our framework uses Qwen3-VL-4B [4] as the backbone $\mathcal{M}$. Unless otherwise specified, all results are obtained without any additional training or parameter updates to the backbone. All TTS comparisons use consistent preprocessing, prompts, answer normalization, and evaluation settings. We set the retry limit R=2 and vary the maximum evidence budget K in the test-time scaling study. Bounding box coordinates are normalized

Table 1: Comparison with representative general-domain and medical vision-language models on OmniMedVQA across eight imaging modalities. Best results are in bold, and second-best results are underlined.

Model	#Params	CT	FP	MRI	OCT	Der	Mic	X-Ray	US	Avg. Acc.
LLaVA-v1.6-34B [33]	34B	50.6	63.4	60.9	68.4	65.7	62.8	74.7	44.5	61.4
Yi-VL-34B [1]	34B	39.8	57.2	51.4	70.5	54.5	61.4	64.2	40.5	54.9
MedGemma [44]	27B	53.8	75.2	65.2	61.6	61.4	60.3	76.4	43.6	62.2
RadFM [51]	14B	33.3	35.0	22.0	31.3	36.3	28.0	31.5	26.1	30.5
LLaVA-v1.6-13B [33]	13B	40.0	43.6	47.4	63.2	58.0	50.5	59.6	42.6	50.6
Med-Flamingo [38]	8B	34.6	33.3	27.5	26.0	28.3	28.1	30.1	33.2	30.2
Qwen3-VL-8B [4]	8B	72.5	76.0	71.7	68.9	79.1	75.0	84.2	56.1	72.9
Lingshu-7B [26]	7B	**78.7**	**88.3**	**84.9**	**90.5**	**81.0**	80.6	83.9	80.0	**83.5**
LLaVA-Med-7B [28]	7B	25.3	48.4	35.9	42.1	45.2	44.0	31.7	83.7	44.5
Qwen-VL-Chat [3]	7B	51.5	45.4	43.9	54.0	55.4	49.5	63.1	33.5	49.5
LLaVA-v1.6-7B [33]	7B	40.1	39.5	54.8	58.4	54.0	48.8	53.3	47.9	49.6
HuatuoGPT-Vision-7B [6]	7B	61.6	80.2	65.1	86.3	71.6	67.4	81.4	**87.4**	75.1
Ours	**4B**	77.5	87.9	74.5	74.9	78.4	**81.5**	**86.8**	82.1	80.5

Table 2: Comparison with representative general-domain and medical vision-language models on four medical VQA benchmarks: PMC-VQA, PathVQA, SLAKE, and VQA-RAD. Our 4B model achieves strong average performance at its scale, demonstrating its generalization ability across diverse medical VQA tasks. Best results are in bold and second-best results are underlined.

Model	#Params	PMC-VQA [58]	PathVQA [14]	SLAKE [32]	VQA-RAD [27]	Avg. Acc.
LLaVA-v1.6-34B [33]	34B	44.4	59.1	67.3	58.6	57.4
Yi-VL-34B [1]	34B	39.5	47.3	58.9	53.0	49.7
MedGemma [44]	27B	37.5	63.1	62.5	71.3	58.6
RadFM [51]	14B	25.9	38.7	34.6	50.6	37.5
LLaVA-v1.6-13B [33]	13B	36.6	51.9	58.9	55.8	50.8
Med-Flamingo [38]	8B	23.3	54.7	43.5	45.4	41.7
LLaVA-Med-7B [28]	7B	24.7	56.8	48.6	51.4	45.4
Qwen3-VL-8B [4]	8B	52.0	**67.5**	69.5	68.5	64.4
Qwen-VL-Chat [3]	7B	36.6	55.1	56.0	47.0	48.9
LLaVA-v1.6-7B [33]	7B	35.5	47.9	57.9	52.6	48.5
HuatuoGPT-Vision-7B [6]	7B	**52.7**	59.9	74.5	63.8	62.7
Ours	**4B**	50.9	64.4	**75.7**	**71.7**	**65.7**

Table 3: Training-free TTS on OmniMedVQA. Best results are in bold.

TTS Strategy	CT	FP	MRI	OCT	Der	Mic	X-Ray	US	Avg. Acc. [95% CI]	#Tokens	p_{McN}	Latency (s)
Single-pass	51.6	85.6	68.4	69.8	74.7	76.2	84.1	65.6	72.0 [71.1,73.0]	562	$< 10^{-4}$	0.0687
Best-of-N (N=6)	64.1	85.9	67.6	70.9	77.3	79.0	86.3	69.3	75.1 [74.1,76.0]	3428	$< 10^{-4}$	0.7088
Majority Voting (N=6)	63.6	84.5	68.1	70.7	77.4	79.4	86.7	68.8	74.9 [74.0,75.8]	3462	$< 10^{-4}$	0.6614
MCTS (R=3, D=1)	50.5	74.8	57.7	65.3	70.9	76.9	80.4	63.0	67.4 [66.4,68.4]	3404	$< 10^{-4}$	2.0228
Ours	**77.5**	**87.9**	**74.5**	**74.9**	**78.4**	**81.5**	**86.8**	**82.1**	**80.5 [79.6,81.3]**	3107	-	1.2835

to $[0, 1000]$, and duplicate evidence is rejected via IoU thresholding at δ=0.7. Experiments are conducted using NVIDIA A100 GPUs.

Benchmarks. We comprehensively evaluate our method on five representative medical VQA benchmarks: OmniMedVQA [15], VQA-RAD [27], SLAKE [32], PathVQA [14], and PMC-VQA [58]. OmniMedVQA is a large-scale benchmark constructed from 73 medical datasets, covering 12 imaging modalities and more than 20 anatomical regions. In our evaluation, we focus on eight modalities, including CT, Fundus Photography (FP), MRI, OCT, Dermoscopy (Der), Microscopy (Mic), X-Ray, and Ultrasound (US), designed to evaluate cross-modality generalization and unified medical visual reasoning. VQA-RAD is a clinician-annotated benchmark focusing on radiology images and clinically grounded question answering. SLAKE

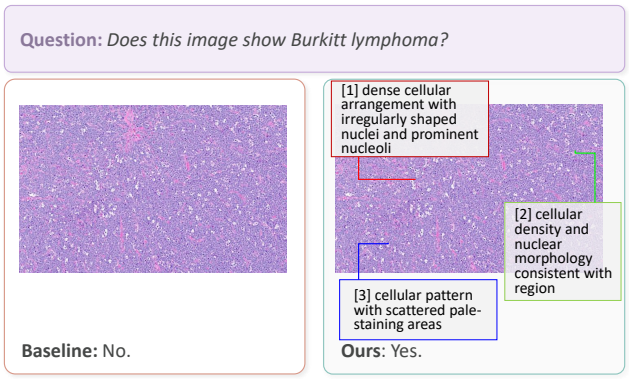


Figure 3: A qualitative example of MedVES. The baseline prediction is incorrect, while MedVES progressively accumulates informative local evidence over multiple evidence-seeking steps and reaches the correct answer.

is a VQA benchmark based on radiology images, with broader body-part coverage and richer question types than earlier datasets. PathVQA targets pathology visual question answering and requires fine-grained understanding of pathology images. PMC-VQA is constructed from figures from PubMed Central articles and evaluates medical visual question answering in the context of biomedical literature. We report accuracy-based results for all benchmarks.

Baselines. We compare our method with a diverse set of representative medical VLMs and general-purpose VLMs spanning diverse parameter scales. The medical-domain models include Lingshu-7B model [26], Med-Flamingo (8B) [38], RadFM (14B) [51], LLaVA-Med-7B [28], HuatuoGPT-Vision (7B/34B) [6], and MedGemma (27B) [44]. We further include strong general-purpose VLMs adapted to medical benchmarks, including LLaVA-v1.6 (7B/13B/34B) [33], Yi-VL-34B [1], Qwen-VL-Chat (7B) [3], and Qwen3-VL (8B) [4]. This selection enables comparison across both domain-specialized and large-scale generalist models under varying capacity regimes.

TTS baselines. We evaluate representative training-free TTS baselines on the same frozen Qwen3-VL-4B backbone. Stronger verifier-assisted methods are complementary, but they introduce auxiliary models, extra supervision, or additional system components, and therefore do not isolate the effect of visual evidence scaling in the training-free setting studied here. **Majority Voting** samples N stochastic candidates and returns the most frequent final answer under self-consistency decoding [50]. **Best-of-N** samples the same N candidates and selects a single response by assigning each candidate a scalar score $s(x, y)$ and returning the highest-scoring sample. In our setting, $s(x, y)$ is defined by the model's own answer confidence, estimated from token log-probabilities when available. **MCTS** performs a UCT-style search over intermediate reasoning traces [9, 23]. In each rollout, UCB1 is used to select a partial reasoning path, one new thought node is expanded, the trace is completed to depth D, and a final answer is produced. The terminal reward is defined by the confidence of the leaf answer when log-probabilities are available; otherwise, we use a rollout-agreement signal as a fallback. In Table 3, we set N=6 for Majority Voting and Best-of-N, and use R=3 rollouts with depth D=1 for MCTS, where R denotes the number of rollouts rather than an explicit branching factor.

4.2 Main Results

OmniMedVQA. Table 1 reports results on OmniMedVQA across eight imaging modalities. Our 4B model achieves an average accuracy of 80.5, demonstrating competitive performance at its parameter scale. The gains are especially notable on CT, FP, MRI, Mic, and X-Ray, suggesting that the proposed reflective evidence-seeking process is particularly effective for modalities that benefit from iterative localization and progressive evidence accumulation.

Four Medical VQA Benchmarks. Table 2 presents results on four medical VQA benchmarks: PMC-VQA, PathVQA, SLAKE, and VQA-RAD. Our model achieves strong average performance at its scale, indicating that the approach generalizes well beyond a single benchmark or modality. The consistent gains across these datasets suggest that the benefit of visual evidence seeking is not limited to one particular task setting, but extends to diverse medical VQA scenarios with different image types, question styles, and reasoning demands.

Qualitative Analysis. Figure 3 presents three representative examples that illustrate how test-time scaling improves medical visual reasoning by progressively accumulating clinically meaningful evidence. In the OCT case, the baseline produces a clearly off-target prediction, while our method focuses on the disrupted hyperreflective retinal layers and the dark central defect across multiple local regions, leading to the correct recognition of a macular hole. In the amyloid fibril example, the model repeatedly inspects both the paired green filaments and the stacked pink structures, collecting complementary cues about the central gap and the internal filament pattern before correctly answering the structural question. In the histopathology case, the model attends to several dense cellular regions characterized by irregular nuclei, prominent nucleoli, and scattered pale-staining areas, which together support the positive identification of Burkitt lymphoma. Across these diverse modalities, the qualitative results suggest that test-time scaling helps the model move beyond a single global guess and instead refine its prediction through step-by-step evidence aggregation, resulting in more accurate and better-grounded answers.

Table 4: Fine-grained ablations on OmniMedVQA.

Setting	Avg. Acc.
W/o evidence listing	80.2
W/o structured output	76.0
W/o explicit localization	78.4
W/o action-conditioned controller	77.6
Ours	**80.5**

Evidence Quality. To assess box relevance, a clinical expert rated 100 model-generated boxes from 51 randomly sampled image-question pairs using a 5-point relevance scale (5, primary evidence; 4, contextual/comparative evidence; 3, benign/normal-region screening; 2, low-value/redundant evidence; and 1, artifact-driven focus). 89.0%

Table 5: Comparison between different framework variants.

Framework Variant	Avg. Acc.	#Tokens
Single-pass	72.0	562
W/o VOS (text-only state)	73.3	2346
W/o cumulative overlay	80.3	4065
W/o adaptive budget	80.8	6148
Ours	80.5	3107

Table 6: Comparison between adaptive and fixed budgets.

Budget Setting	Avg. Acc.	#Tokens
Fixed K=1	79.2	1155
Fixed K=2	79.6	2192
Fixed K=3	80.1	3440
Fixed K=4	80.6	4780
Fixed K=5	80.8	6148
Ours	80.5	3107

scored 4-5 (mean 4.65 ± 0.87), whereas 6.0% scored 1-2. Clinical focus rates (scores 4-5) were 87.3% for global and 91.9% for localized/lesion-specific questions. The relevance scores showed no significant difference (Mann-Whitney $U = 321.0$, $p = 0.884$), indicating that the predicted boxes generally captured clinically meaningful evidence.

Inference Time Scaling. We show that the proposed visual evidence scaling method improves model capability at inference time when given a larger token budget. Figure 2 shows that our method benefits substantially from inference-time scaling. As the budget increases, performance improves steadily, confirming that additional evidence-seeking steps can translate into better predictions. However, without a mechanism to limit computation, these gains come at the cost of increasing token usage. Our adaptive early-exit strategy lies near the top of the efficiency-performance trade-off: it achieves an accuracy of 80.5 with only 3,107 tokens, clearly outperforming smaller fixed-budget settings while remaining close to the best large-budget results and requiring far less computation.

Cross-Backbone Generalization. We also observe positive transfer of MedVES to another 4B backbone [44]. Detailed results are provided in the supplementary material due to space limitations.

4.3 Ablation Studies

Effect of TTS Strategy. To examine whether increasing language-space computation alone is sufficient for effective TTS in medical VQA, we compare MedVES with Best-of-N, Majority Voting, and MCTS under the same frozen backbone. As shown in Table 3, MedVES achieves the highest average accuracy and significantly outperforms single-pass inference and all three language-space TTS baselines under paired McNemar tests [37], while using fewer tokens than the three TTS baselines. These results support allocating additional inference-time compute to iterative visual evidence seeking rather than language-space sampling or search alone.

Effect of Framework Design. Table 5 evaluates the contribution of three core design choices in MedVES. Replacing the VOS with a textual evidence state markedly lowers performance, indicating that visually and spatially grounded memory is more effective than textual carry-over for iterative evidence seeking. Removing cumulative overlay and instead appending per-step overlay images to the dialogue history preserves similar accuracy but significantly increases token usage, showing that consolidating accepted evidence into a single persistent visual state is important for both effective reasoning and efficient inference. Finally, replacing the adaptive evidence budget with a fixed budget of $K = 5$ slightly improves accuracy, but at a significantly higher token cost, confirming the expected scaling benefit of additional evidence steps while showing that adaptive early exit provides a more favorable efficiency-performance trade-off. Furthermore, Table 4 reports finer-grained ablations: w/o evidence listing removes accumulated `OBS` entries from the final-answer prompt; w/o structured output relaxes the `BBOX/OBS/NEXT` format while retaining basic box parsing; w/o explicit localization replaces box-based VOS updates with full-image textual observations; and w/o action-conditioned controller removes `NEXT` and uses generic evidence-seeking queries. MedVES primarily benefits from structured, spatially grounded, and action-conditioned evidence seeking, with only marginal gains from explicit evidence listing.

Effect of Adaptive Evidence Budget. Table 6 compares adaptive stopping with fixed evidence budgets. Increasing the budget K monotonically improves performance, but incurs higher token and latency costs. Adaptive MedVES achieves 80.5 accuracy with 3107 tokens, whereas fixed K=4 improves accuracy by only 0.1 points while using 53.8% more tokens and 2.05× the latency, demonstrating a favorable test-time accuracy-cost trade-off. The first evidence step provides the largest marginal gain, as question-conditioned localization often captures sparse, discriminative evidence; subsequent steps mainly refine, verify, or contextualize it. Among 50 sampled K=1/2 early-stop failures, expert review found that only one lacked a primary-evidence box, suggesting that most residual errors arise from evidence interpretation rather than acquisition.

5 Conclusion

We present MedVES, a training-free test-time scaling framework for medical VQA. We identify a fundamental limitation of existing TTS approaches: generating more text tokens merely elaborates on the model's initial visual impression rather than uncovering new visual evidence. MedVES addresses this by reframing inference-time scaling as iterative visual evidence seeking, where a frozen VLM progressively attends to complementary image regions before producing an answer. Its VOS externalizes accumulated evidence as bounding box overlays on the original image, providing a spatially grounded and globally consistent state, while an adaptive evidence budget allocates compute according to question complexity. Across five medical VQA benchmarks, MedVES improves frozen VLM inference without additional training and achieves competitive performance. More broadly, our results suggest that visual evidence seeking is an effective way to allocate additional test-time compute for medical VQA. We hope that MedVES opens a new direction for inference-time scaling beyond purely text-based reasoning.

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